1. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52

(a) $\left[\text{Mean} = \sum_{i=1}^{n} \frac{w_i x_i}{w_i}\right]$ $\frac{1}{N} \sum_{i=1}^{n} \frac{w_i x_i}{w_i}$

$= \frac{809}{27}$

$= 29.96 = 30$

Median $= \frac{n+1}{2}$

$= \frac{27+1}{2}$

$= \underline{14}^{th} = 25$

Median $= \underline{\underline{25}}$

b] Mode and modality

frequency

25 and 35 are repeating for 4 times it is bimodel

(c) Midran = $\frac{minimum + maximum}{2}$

$= \frac{13+70}{2}$

$= 41.5$

Midrange $= 41.5$

d) First Quartile (Q_1) and Third Quartile Q_3

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25

Q_1 and Q_3 Lower half

Middle value = 20

$Q_1 = 20$

30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

upper half

middle value = 35

$Q_3 = 35$

IQR = 15

$Q_3 - Q_1$

e) Five Number summary

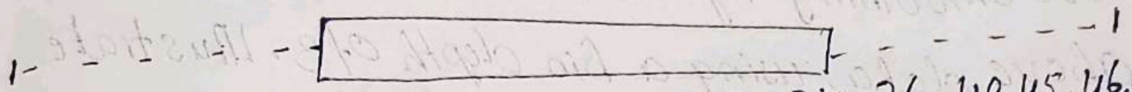
min, Q_1, median, Q_3, max

- minimum = 13
- $Q_1 = 20$
- median = 25
- $Q_3 = 35$
- maximum = 70

f. Boxplot of the data

13, 15, 16, 19, 20, 21, 22, 25, 30, 33, 35, 36, 40

45, 46, 52, 70



13, 15, 16, 19, 20, 21, 22, 25, 30, 33, 35, 36, 40, 45, 46, 52, 70

2. Given the following measurements for the variable

age: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28

standarize the variable by the following

compute the mean absolute deviation of age

Given data

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Mean $\bar{x} = \frac{330}{10} = 33$

Age $(x - \bar{x})$

18 - 15

22 - 11

25 - 8

42 - 9

28 - 5

43 - 10

33 - 0

35 - 2

56 - 23

28 - 5

$MAD = \frac{88}{10} = 8.8$

3. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order)

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Use smoothing by bin means to smooth the above data, using a bin depth of 3. Illustrate your steps.

Total values = 27

no. bin depth = 3

Total no. of bins = $\frac{27}{3}$ = 9 bins required

Bin	Values	Bin mean	Smoothed value
B_1	13, 15, 16	14.67	14.67, 14.67, 14.6
B_2	16, 19, 20	18.33	18.33, 18.33, 18.
B_3	20, 21, 22	21.00	21.00, 21.00, 21.0
B_4	22, 25, 25	24.00	24.00, 24.00, 24
B_5	25, 25, 30	26.67	26.67, 26.67, 26.6
B_6	33, 33, 35	33.67	33.67, 33.67, 33.6
B_7	35, 35, 35	35.00	35.00, 35.00, 35.
B_8	36, 40, 45	40.33	40.33, 40.33, 40.3
B_9	46, 52, 70	56.00	56.00, 56.00, 56.00

4. Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000 (a) min-max normalization by setting min = 0 and max = 1 (b) z-score normalization

200, 300, 400, 600, 1000

(a) min-max

$$x' = \frac{x - Min}{Max - Min}$$

min = 200

max = 1000

Value	Normalizele
200	(200-200)/800 = 0.00
300	$\frac{300-200}{800}$ = 0.125
400	(400-200)/800 = 0.250
600	(600-200)/800 = 0.500
1000	(1000-200)/800 = 1.000

(b) $z = \frac{x - \mu}{\sigma}$

$$\mu = \frac{200 + 300 + 400 + 500 + 1000}{5} = 500$$

Value	$(x - \mu)$	$(x - \mu)^2$
200	-300	1.061
300	-200	0.707
400	-100	0.354
600	-400	0.354
1000	-500	1.768

5. Suppose a group of 12 sales price reco[rds] has been sorted as follows: 5, 10, 11, 13, 15, 50, 55, 72, 92, 204, 215. paritition them into th[ree] ... following methods. (a)